# Multi-Agent Visual SLAM for 3D Coastline Reconstruction: Overcoming Synchronization and Dynamic Environment Challenges

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***Abstract*—3D coastline reconstruction presents a significant challenge due to the complex and dynamic environment of coastal areas, characterized by fluctuating sea levels, vegetation, and irregular terrain. Visual Simultaneous Localization and Mapping (V-SLAM) technology offers a promising approach to addressing these challenges by enabling real-time mapping and pose estimation. Conventional V-SLAM methods, which are based on single-agent, are inadequate to meet the complex requirements of such environments. Consequently, multi-agent V-SLAM has emerged as a promising alternative; however, its implementation remains challenged by several unresolved issues. This paper examines the difficulties associated with multi-agent V-SLAM concerning synchronization and points out potential solutions to enhance its efficacy. Additionally, 3D coastline reconstruction is used as a case study to demonstrate the challenges encountered in real-world applications of multi-agent V-SLAM and to propose strategies for overcoming these obstacles.**

***Index Terms*—V-SLAM, Multi-Agent SLAM, 3d Coastline Reconstruction**

## I. Introduction

### A. Purpose of the Review

Reconstructing coastlines in 3D presents a significant challenge due to the dynamic and multifaceted nature of coastal environments, which are subject to continuous changes influenced by tidal fluctuations, vegetation growth, and shifting topography [1]. The generation of precise 3D coastline models is critical for a range of applications, including environmental monitoring, coastal erosion analysis, and the planning and maintenance of infrastructure. Despite their widespread use, conventional single-agent Visual Simultaneous Localization and Mapping (V-SLAM) systems are often constrained by their limited spatial coverage and inability to effectively manage the complexity inherent in coastal environments [2]. To overcome these limitations, multi-agent SLAM systems have been introduced as a more robust and scalable approach. By deploying multiple autonomous agents that cooperate to simultaneously localize themselves and map their surroundings, multi-agent SLAM significantly enhances spatial coverage, improves mapping accuracy, and increases the overall robustness of mapping efforts in large, dynamic environments such as coastlines [3].

This study investigates the application of multi-agent V-SLAM technology in 3D coastline reconstruction, emphasizing the unique technical challenges posed by coastal environments and the potential of multi-agent systems to overcome these obstacles. The paper examines critical issues such as the dynamic characteristics of coastal regions, occlusion management, and the synchronization of multiple agents in large-scale environments. By leveraging recent advances in V-SLAM, and deep-learning-based approaches, multi-agent SLAM systems offer the transformative potential for increasing the precision, efficiency, and scalability of coastline mapping, with significant implications for environmental monitoring, disaster response, and the development of energy infrastructure.

### B. Background on V-SLAM Systems

V-SLAM is a specialized subset of the broader SLAM technology, which can be categorized into two main approaches: LiDAR SLAM [4], which utilizes laser radar [5], and V-SLAM, which relies on camera-based input. While LiDAR SLAM is highly effective in obtaining accurate depth information due to its laser-based sensing capabilities, it cannot capture detailed RGB and semantic information necessary for a comprehensive understanding of the environment [6]. This limitation reduces its applicability for tasks requiring rich contextual data. In contrast, V-SLAM, which utilizes visual data from cameras, provides a more versatile and expansive framework, making it the preferred choice for 3D modeling in large-scale environments where detailed visual and semantic information is crucial.

V-SLAM Technology is a computational framework that enables a system, typically equipped with a monocular [7], stereo, or RGB-D camera [8], to concurrently estimate its own pose while reconstructing the surrounding environment in real-time. The architecture of Visual SLAM is conventionally

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organized into three core components: **Tracking**, **Mapping**, and **Loop Closing**. Each of these components plays a crucial role in ensuring the accuracy and robustness of the SLAM process [7]. The overall architecture of V-SLAM is presented in Fig. 1.

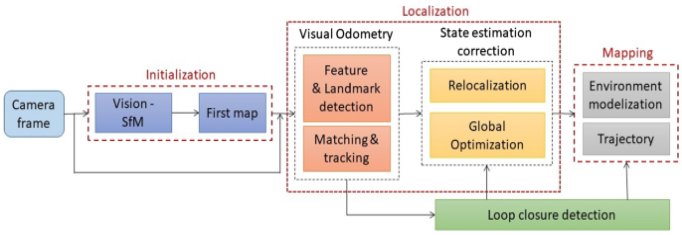


Fig. 1. The overall architecture of V-SLAM [9].

Single-agent V-SLAM technology faces significant challenges in large and complex terrains. Its narrow field of view and limited mobility make it difficult to survey extensive and varied landscapes effectively. Navigating these terrains requires frequent movement and repositioning, which is time-consuming and can lead to data loss due to obstacles like vegetation, cliffs, buildings, and uneven ground [10]. Additionally, the single viewpoint restricts the ability to capture the multiple perspectives needed for accurate 3D mapping of varied features such as rugged landscapes and changing elevation [11]. These limitations often result in incomplete or inconsistent data, making it harder to create precise 3D models, which are critical for understanding and managing complex terrains effectively.

### C. Objectives and Structure

This review aims to assess the feasibility and effectiveness of Multi-Agent V-SLAM in overcoming the inherent challenges of single-agent systems, particularly in the context of 3D coastline reconstruction for energy applications. The review is structured as follows: the subsequent section provides an overview of Multi-Agent V-SLAM systems, detailing existing challenges, and potential solutions. This is followed by an in-depth discussion on the application of Multi-Agent V-SLAM to 3D coastline reconstruction, focusing on the specific technical challenges encountered in coastal environments and the strategies with the potential to address them. The review summarizes the main findings and delineates directions for future research and practical advancements.

## II. Overview of Multi-Agent SLAM System

Multi-Agent V-SLAM extends conventional Single-Agent V-SLAM by employing multiple agents, including drones or robots, to collaboratively map and localize within an environment, thereby addressing the scale and dynamic environment limitations inherent in single-agent systems. This multi-agent approach relies on decentralized coordination, where each agent independently processes its visual and sensor data to construct local maps and estimate poses, while simultaneously maintaining real-time communication with other agents to share information and integrate local maps into a cohesive global map.

Nevertheless, achieving synchronization in multi-agent V-SLAM presents a significant challenge, requiring the resolution of multiple issues, including inconsistencies in data fusion and disparities in sensing and processing times among agents [3]. Visual place recognition [12] plays a key role in addressing these challenges by helping to align the maps and trajectories of different agents. By recognizing when multiple agents observe the same location from different perspectives, visual place recognition can correct drift in position estimates and ensure a consistent global map, thus facilitating effective synchronization of spatial data across the agents.

### A. Visual Place Recognition: Overview and Importance

Visual place recognition involves identifying previously visited locations using visual information and is also known as inter-agent loop detection. Within the context of multi-agent V-SLAM, visual place recognition is essential for synchronizing the trajectories of multiple agents by identifying overlaps in their observations.

While single-agent visual place recognition involves recognizing when a single agent returns to a previously visited location, multi-agent visual place recognition requires detecting shared observations made by different agents. Effective visual place recognition is essential for collaborative mapping and reducing redundancy in data collection among multiple agents.

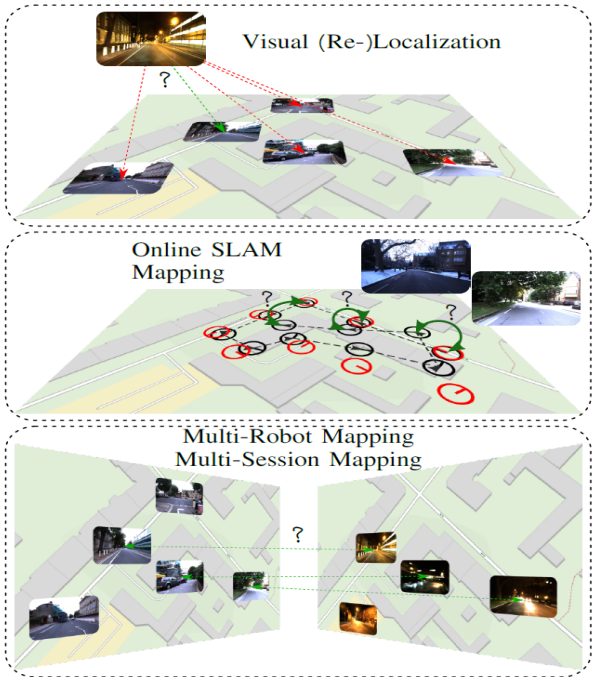


Fig. 2. Visual Place Recognition

### B. Methods for Visual Place Recognition

*1) Conventional Methods:* Feature-Based Approaches: Conventional methods for visual place recognition rely on

detecting and matching visual features across different agents' observations. Techniques such as SIFT [13], SURF [14], and ORB [7] extract key points and descriptors from images, enabling recognition of places that remain relatively stable under changes in scale, rotation, and illumination.

Bag of Words (BoW) Models [7] build on these feature-based approaches by creating a visual vocabulary from local features. Images are represented as histograms of these visual words, allowing for efficient comparison and recognition of previously visited locations across agents. BoW models are effective in environments where scenes do not change drastically over time but face challenges in dynamic settings due to their reliance on handcrafted features and lack of robustness to environmental variations. Fig. 3 visualizes BOW.

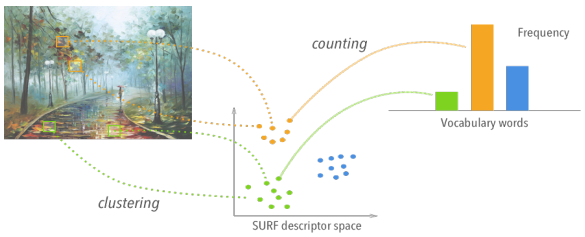


Fig. 3. Bag of Words(BOW) [15]

*2) Deep Learning-Based Approaches:* Deep learning techniques, particularly those involving neural networks, have transformed visual place recognition by enabling automatic feature learning from data. Convolutional Neural Networks (CNNs) [16] have been extensively used to extract high-level features from images, enhancing place recognition by learning complex patterns directly from raw data, thereby improving robustness to variations in lighting, scale, and perspective. More recently, transformer-based models [17], which employ self-attention mechanisms to capture long-range dependencies in visual data, have emerged as a promising development in deep learning-based loop closure detection. These models potentially offer superior capabilities in understanding complex scenes and detecting inter-agent loop closures across diverse environments.

The Transformer-based study, EffoVPR [18], introduces a novel approach that leverages features extracted directly from the self-attention layers of a Vision Transformer (ViT) [19]to perform zero-shot re-ranking. This method circumvents the need for external pooling layers or specialized aggregation techniques, enhancing its robustness and generalizability compared to existing methods. In the Visual Place Recognition challenge of the Pittsburgh-30k-test dataset [20], EffoVPR achieved a leading performance with a Recall@1 score of 93.9, the highest reported to date. Fig. 4 shows the overall architecture of EffoVPR.

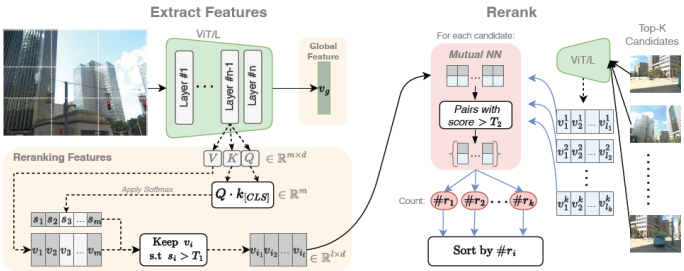


Fig. 4. EffoVPR Architecture: On the left, the inference process involves identifying the nearest neighbors of the query by utilizing the [CLS] token as the global representation of each image ($v_g$). In the second re-ranking stage, intermediate value features $V$ are extracted (indicated by dashed lines) and filtered using a predefined threshold $T_1$, based on a score calculated from the partial attention maps $S$. On the right, the top-$K$ candidates from the initial stage are further re-ranked by evaluating the number of strongly connected mutual nearest neighbors (MNN) whose scores exceed a specified threshold (denoted as $T_2$) [18].

Similarly, ProGEO [21] demonstrates strong performance on the same dataset. The authors present a two-stage training strategy that utilizes the multi-modal capabilities of CLIP (Contrastive Language-Image Pretraining) [22], a model conventionally designed to integrate visual and textual data. In the first stage, ProGEO employs learnable text prompts to generate abstract descriptions for each geographic image feature, capitalizing on the descriptive capacity of CLIP. In the second stage, these dynamically generated text prompts are utilized to train an image encoder, enabling it to learn more generalizable visual features. This approach effectively addresses the challenges associated with using multi-modal visual models for geo-localization, particularly in cases where images lack explicit textual descriptions, thereby enhancing the robustness and accuracy of location identification. ProGEO achieved a Recall@5 score of 98.3 on the Pittsburgh-30k-test Benchmark, which currently represents the highest score in this category.

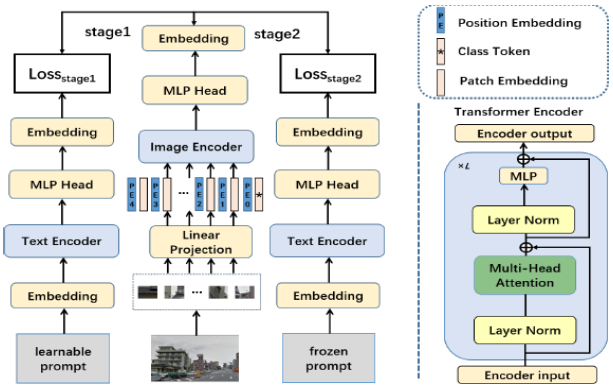


Fig. 5. The overall architecture of ProGEO model with a ViT image encoder backbone [21].

## C. Future Directions for Multi-Agent V-SLAM

Future research on Multi-Agent V-SLAM should focus on optimizing hardware requirements [23] and enhancing model inference speed to ensure practical deployment in real-world

scenarios. Although recent advancements have significantly improved visual place recognition compared to conventional techniques, these benefits are often constrained by the high computational demands and the specialized hardware needed to run such systems efficiently. A critical direction for future work involves developing algorithms that can operate effectively on lower-cost, more accessible hardware without compromising accuracy or robustness. In summary, while the deep learning-based method offers promising improvements over conventional approaches, its future lies in balancing performance gains with practical considerations including cost, hardware accessibility, and operational speed.

## III. Challenges in Multi-Agent 3D Coastline Reconstruction

### A. Overview of Multi-Agent 3D Coastline Reconstruction

3D coastline reconstruction is a challenging task that aims to create detailed and accurate models of coastal environments for various applications, such as monitoring coastal erosion, assessing habitat changes, managing disaster response, and planning infrastructure development. This task involves acquiring high-resolution spatial data across diverse coastal zones, including cliffs, beaches, dunes, and intertidal areas, using multiple autonomous agents equipped with sensors. These agents must navigate complex and dynamic environments to capture overlapping images or point clouds while performing real-time localization to construct coherent 3D maps [24]. The process also requires integrating data collected from multiple agents, which involves synchronizing data streams, merging overlapping datasets, and accounting for dynamic factors such as waves, tides, and moving vegetation. Additionally, the task necessitates continuous updating and refinement of the 3D models to reflect changes in the coastal environment over time, ensuring accuracy and relevance for decision-making.

### B. Dynamic Environment and Occlusion Challenges

The dynamic nature of coastlines, characterized by flowing water, fluctuating sea levels, and tidal variations, poses significant challenges for multi-agent 3D reconstruction using V-SLAM systems. Unlike static environments, coastlines are subject to constant changes that can cause frequent disruptions in the mapping process, leading to reduced accuracy and stability in the resulting 3D models [25]. These dynamic elements, such as waves and tides, cause occlusion by obscuring important shoreline features, making it difficult for V-SLAM algorithms to maintain consistent accuracy. The movement of water can interfere with the system's ability to detect and map critical areas, causing it to misinterpret or fail to recognize important features. This combination of continuous environmental changes and occlusion significantly reduces the robustness and reliability of the SLAM-based 3D reconstruction process.

### C. Potential Solution

To address these combined challenges, a motion frame analysis module is proposed to enhance the V-SLAM system's ability to handle dynamic environments. This module aims to predict ocean level movements and dynamics in real time, enabling the V-SLAM system to differentiate between static land features and the constantly changing water surfaces. By focusing on stable elements and compensating for fluid movement, the motion frame analysis module can improve feature recognition and mapping accuracy.

Liu et al. present a novel semi-weakly supervised learning framework to address the challenging task of predicting kinematic motions of 3D objects [26]. The proposed method utilizes a graph neural network (GNN) [27] to establish a correspondence between hierarchical part-level segmentation and motion parameters, subsequently refining these predictions based on geometric alignment. Initially, the GNN is trained on the PartNet-Mobility dataset [28] to learn the mapping between semantic segmentation hierarchies and mobility information. This model is then applied to the PartNet dataset to generate pseudo-labeled data, enabling the development of a semi-weakly supervised learning model that can effectively learn from both labeled and pseudo-labeled data. Experimental results indicate that augmenting the training data in this manner significantly enhances the model's performance in predicting kinematic motions, especially when applied to partial 3D scans.

A primary application of object motion prediction is in the field of self-driving. The research introduces MotionNet [29], a novel deep-learning model designed for autonomous vehicles that simultaneously performs object perception and motion prediction from 3D point clouds. MotionNet utilizes a sequence of LiDAR sweeps to generate a Bird's Eye View (BEV) map [30], which encodes both the object categories and their motion information in each grid cell. The backbone of MotionNet is a unique spatio-temporal pyramid network (STPN) that hierarchically extracts spatial and temporal features. The authors introduce novel spatial and temporal consistency losses to enhance the smoothness and accuracy of predictions across time and space. These losses ensure the model's predictions remain consistent over time, improving the reliability of motion forecasts. The authors validate MotionNet's performance using the large-scale nuScenes dataset [31], demonstrating its superior performance over existing state-of-the-art methods in terms of accuracy and efficiency. While this study is currently constrained to the bird's-eye view (BEV) perspective and does not extend to a three-dimensional coordinate system, the conceptual framework and methodologies presented offer valuable insights and merit further consideration for future research applications.

## IV. Conclusion

The literature review highlights several critical deficiencies in the current V-SLAM systems for constructing 3D coastlines. Key challenges include ineffective visual place recognition and the problem of dynamic occlusion where moving objects obscure significant portions of the scene. To address these limitations, integrating deep learning techniques has been proposed as a potential solution. However, further

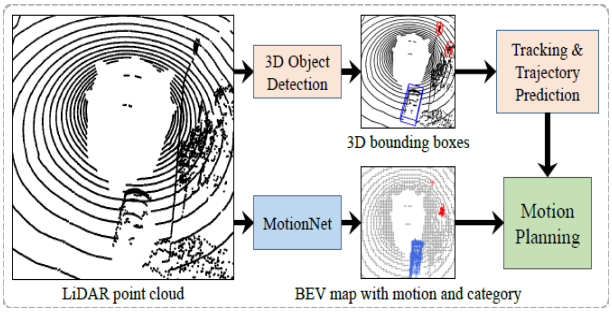


Fig. 6. MotionNet is a bird's-eye-view (BEV) map-based system that integrates perception and motion prediction tasks without relying on conventional bounding boxes, where the color represents the different category of object and the arrow denotes the future displacement [29].

research is necessary to integrate these deep learning-based solutions cohesively into V-SLAM frameworks, considering the performance constraints of current hardware and model architectures. Future research and practice implications are vast: researchers must explore new methodologies for seamless integration, and developers need to optimize algorithms for real-time execution. Enhancing V-SLAM systems with deep learning could significantly advance 3D coastline modeling by providing more reliable and precise mapping solutions in dynamic coastal environments, ultimately benefiting environmental monitoring, coastal management, and disaster response efforts.

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